**Accuracy** $S_{acc}$

**Adaptability** $S_{adapt}$

**Structure** $S_{struct}$

**Population Fidelity** Score

**Center alignment** $S_{center}$

**All retained cells**

as released

German

Mexican

distribution-matched

subgroup-matched

**German cells only**

as released

German

distribution-matched

**Mexican cells only**

as released

Mexican

distribution-matched

**United States cells only**

as released

distribution-matched

subgroup-matched

0 0.5 1

Score (0 to 1, higher is better)

Gemma-4-31B-it

Gemma-4-E4B-it

Qwen3-VL-8B-Thinking

Qwen3-VL-2B-Thinking

Llama-3.2-3B

Muse-Glimmer-30B

hollow: as released

solid: German

left half: distribution-matched

bottom half: Mexican

right half: subgroup-matched

95% CI across runs